Here is an illustration of how to construct a probability law starting from some common sense assumptions about a model.

Illustration 12.6

(1) Let $S=\{1,2,3\}$. Suppose $\mathscr{P}(S)$ is the power set of S, and $P(A)=\dfrac{n(A)}{n(S)}$.

Then $P(\{1\})=\dfrac{1}{3}$, $P(\{2\})=\dfrac{1}{3}$, and $P(\{3\})=\dfrac{1}{3}$,

satisfies axioms of probability $[P_1]$,$[P_2]$, and $[P_3]$. Here all the outcomes are equally likely.

(2) Let $S=\{1,2,3\}$. Suppose $\mathscr{P}(S)$ is the power set of S,

If the probability $P(A)$, of an event A of S is defined as the sum of the probabilities of the points in A,

then $P(\{1\})=\dfrac{1}{2}$, $P(\{2\})=\dfrac{1}{4}$, $P(\{3\})=\dfrac{1}{4}$,

satisfy the axioms of probability $[P_1]$,$[P_2]$, and $[P_3]$.

(3) Let $S=\{1,2,3\}$ and $\mathscr{P}(S)$ is the power set of S. If the probability $P(A)$, of an event A of S is defined as the sum of the probabilities of the points in A,

then $P(\{1\})=0$, $P(\{2\})=\dfrac{1}{\sqrt{2}}$, and $P(\{3\})=1-\dfrac{1}{\sqrt{2}}$,

satisfy the above axioms $[P_1]$,$[P_2]$, and $[P_3]$.

In (2) and (3), the outcomes are not equally likely.

Note 12.2

Irrational numbers also can act as probabilities.

Classroom Activity: Each student to flip a coin 10 times,

Calculate: $p=\dfrac{\text{Number of times heads occur}}{10}$

Find the cumulative ratio of heads to tosses. As number of tosses increases $p\to\dfrac{1}{2}$

Example 12.1

If an experiment has exactly the three possible mutually exclusive outcomes A, B, and C, check in each case whether the assignment of probability is permissible.

(i) $P(A)=\dfrac{4}{7}$, $P(B)=\dfrac{1}{7}$, $P(C)=\dfrac{2}{7}$.

(ii) $P(A)=\dfrac{2}{5}$, $P(B)=\dfrac{1}{5}$, $P(C)=\dfrac{3}{5}$.

(iii) $P(A)=0.3$, $P(B)=0.9$, $P(C)=-0.2$.

(iv) $P(A)=\dfrac{1}{\sqrt{3}}$, $P(B)=1-\dfrac{1}{\sqrt{3}}$, $P(C)=0$.

(v) $P(A)=0.421$, $P(B)=0.527$ $P(C)=0.042$.

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